

Homework Assignment 2A

PWM Feedback Loops

- In Cadence Virtuoso create with elements from **analogLib** and **ahdlLib** a 1st order feedback loop around a PWM generator running at 1MHz using an active RC-integrator with $R_{FB} = 100k\Omega$. The output V_{PWM} switching between $\pm 10V$ supply. For the integrator use **limiting_diffamp** from **ahdlLib** and limit the output swing to $\pm 2V$. To model the comparator in the loop use a cascade of **subtractor** and **hysteresis** cells from **ahdlLib** instead of the **comparator** cell. Connect the LC-filter and 4Ω load from Homework Assignment 1A.
 - calculate the value of the integrator capacitor C_{INT} such that the loop fulfills the large-signal stability criterion
 - inject a 1kHz sinusoidal current input signal in the virtual ground of the loop
 - determine the maximum current amplitude $I_{IN,MAX}$ the loop can handle without clipping
 - inject a 5kHz sinusoidal voltage error signal between the switching node V_{PWM} and the feedback resistor R_{FB} and determine the error suppression across the load.
 - calculate the loopgain @ 5kHz and check if this matches the suppression of the error signal (tip: use **spectrumMeasurement** expression to determine the signal power)
- Add a second RC-integrator to create a 2nd order feedback loop. Use the second integrator to generate the triangle reference by injecting a square wave current signal in the virtual ground node of the second integrator
 - calculate an appropriate value for the zero in the loop transfer
 - inject a 5kHz sinusoidal voltage error signal between the switching node V_{PWM} and the feedback resistor R_{FB} and determine the error suppression across the load.
 - calculate the loopgain @ 5kHz and check if this matches the suppression of the error signal
- drive both the 1st order and 2nd order loop into clipping with a signal amplitude of FS+10%.
 - compare the signals after the LC-filter in both cases. describe and explain your observations

Homework Assignment 2B

PWM Feedback Loops

- In Cadence Virtuoso create with elements from **analogLib** and **ahdlLib** a **Direct PWM** 1st order feedback loop running at 600kHz using an active RC-integrator with **$R_{FB} = 100k\Omega$** . The output V_{PWM} switches between $\pm 12V$ supply. For the integrator use **limiting_diffamp** from **ahdlLib** and limit the output swing to $\pm 2V$. To model the comparator in the loop use a **hysteresis** cell from **ahdlLib** instead of the **comparator** cell. Reuse the LC-filter and 8Ω load and the double-sided PWM generator from Homework Assignment 1B. Use a **vccs** (voltage-controlled current source) from **analogLib** to convert the PWM signal into a PWM current I_{PWM} that can be injected into the virtual ground.
 - determine the amplitude of the PWM input current amplitude to get a modulation index gain of 1.6
 - determine the input signal amplitude relative to FS to drive the output to near clipping
 - determine the value of the integrator capacitor C_{INT} such that the integrator output swing is limited to $\pm 2V$.
 - run a simulation with a 0.1FS @ 1kHz input signal.
 - reduce the supply voltage (V_{PWM} amplitude) to $\pm 8V$ and repeat the simulation. Describe and explain your observations
 - re-determine the value of the integrator capacitor C_{INT} such that the integrator output swing is again limited to $\pm 2V$.
- Inject a 5kHz sinusoidal voltage error signal between the switching node V_{PWM} and the feedback resistor R_{FB}
 - determine the error suppression across the load for supply voltages of $\pm 8V$ and $\pm 12V$. describe and explain your observations (tip: use **spectrumMeasurement** expression to determine the signal power)
 - calculate the loopgain @ 5kHz for both cases and check if this matches the suppression of the error signal

Homework Assignment 2C

PWM Feedback Loops

1. In Cadence Virtuoso create an **ideal delay-based** self-oscillating PWM feedback loop using **limiting_diffamp** & **hysteresis** from ahdLib
 - make an active-RC integrator around the **limiting_diffamp** from **ahdLib** and limit the output swing to $\pm 2V$. Use **100k Ω** as feedback resistor
 - set the parameters of the hysteresis so that the hysteresis window is 0 and the output switches between -5V and +5V
 - determine the delay of the hysteresis block so that the switching frequency is 256kHz
 - determine the value of the integrator capacitor C_{INT} such that the integrator output swing is limited to $\pm 2V$.
 - attach the filter and load from Homework 1C to the output
2. Add a signal input to the loop by connecting a signal voltage source with a resistor to the virtual ground node of the integrator
 - run a series of transient simulations with a DC-input voltage ranging from -0.9 [FS] to 0.9 [FS] (remember to set **reltol** to an appropriate value)
 - make a plot of PWM frequency versus input signal level: describe and explain your observations.
 - run a simulation with a 1kHz sinwave input with amplitude 0.1 [FS]
 - make a frequency spectrum plot of the signal after the LC-filter: describe and explain your observations
3. Add error voltage source to the output of the hysteresis block (but before the feedback point)
 - run a series of simulations with a sinewave **error** signal with frequencies {1, 2, 4, 8, 16} [kHz] and amplitude 0.1V
 - plot the frequency spectra of the signal after the LC-filter: describe and explain your observations
 - make a plot of signal power versus frequency (tip: use **spectrumMeasurement** expression) how is this plot related to the error transfer function (ETF) ?
 - extrapolate the plot to determine the **unity-gain frequency** of the loop transfer of the delay-based self-oscillating loop: does this match your expectation?
4. Modify the parameters of the hysteresis block to transform the loop into an **ideal hysteretic** self-oscillating feedback loop
 - set the delay of the hysteresis block to 0 and determine the hysteresis window such that the switching frequency is again 256kHz
 - repeat steps 2 and 3 and compare the results to the delay-based loop: describe and explain your observations